

freeiv: Instrument-free estimation of a linear structural model with an endogenous regressor

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Abstract. `freeiv` estimates the coefficient on an endogenous regressor when no external instrument is available, for the linear triangular model in which the endogeneity is generated by a single latent confounder. The command is organised around a result that costs no distributional assumption: under a single restriction on the confounder’s two loadings, the coefficient lies between half the ordinary least-squares slope and that slope itself, and that interval is exactly the region in which the implied variances are non-negative. Every point estimator then buys identification with one further assumption, so `freeiv` reports the routes side by side against the interval, together with the diagnostic that says whether each one’s assumption is carried by the data. Nine routes belong to the maintained model — closed forms at second and third order, a quantile-based extrapolation, and two generalized method of moments estimators on the moments of orders two, three, and four, each with a Hansen J statistic — and seven more, among them the estimators of Lewbel (2012, *Journal of Business and Economic Statistics* 30: 67–80), of Lewbel, Schennach, and Zhang (2024, *Journal of Business and Economic Statistics* 42: 14–25), and of Oster (2019, *Journal of Business and Economic Statistics* 37: 187–204), are computed on the same data for comparison. Two endogenous regressors loading on one confounder switch the command to a two-indicator model in which the confounder’s direct loading is free and estimated, which is what makes the maintained restriction itself testable. Companion commands report what the data can carry before any estimation, what a given coefficient implies about the unobservables, and a full diagnostic table.

Keywords: `freeiv`, `freeivmenu`, `freeivdiag`, `freeivtest`, `freeivreport`, endogeneity, latent confounder, scale consistency, partial identification, higher-order moment, instrument-free identification

1 Introduction

The applied researcher who suspects that a regressor is endogenous and has no credible instrument has, in practice, three options. Report the least-squares coefficient and discuss the direction of the bias. Find an instrument whose exclusion restriction is defensible, which is rarely possible and never testable. Or turn to one of the instrument-free estimators that identify the coefficient from features of the data other than an exclusion restriction: heteroscedasticity, higher-order moments, a functional form for the dependence between the disturbances.

The third route has produced a substantial literature, and it has an uncomfortable property. Each of these estimators identifies the parameter inside its own maintained model, and the maintained models differ. Applied to the same dataset they give different numbers, and the difference is not sampling noise: it is the disagreement between the assumptions. A reader shown one of them has no way to tell whether the assumption behind it holds.

`freeiv` is built around that difficulty rather than around any one estimator. It takes the maintained model seriously enough to ask what can be said before any distributional assumption is added, reports the answer as an interval, and then shows every point estimator against that interval together with the statistic that says whether its own assumption is carried by these data. An estimate that falls outside the interval is not a large estimate; it is an infeasible one, implying a negative variance under the maintained model, and the command says so instead of printing it without comment.

The article is in two halves. Section 2 presents the models: what each one assumes, what it identifies, and how the command estimates it. There are no proofs here — they are in the papers cited at each point — and the notation is kept to what a practitioner needs to decide whether a route applies to the problem at hand. Section 3 onwards presents the command itself: syntax, options, how to read the output, what is stored, and worked examples on simulated data where the truth is known and on the Card (1995) extract where it is not.

2 The models

2.1 The common setting

Throughout, Y_1 is the outcome, Y_2 the endogenous regressor, X a vector of exogenous controls including a constant, and γ_1 the coefficient of interest:

$$Y_1 = X'\beta_1 + \gamma_1 Y_2 + \varepsilon_1, \quad \varepsilon_1 = \alpha_1 U + V_1, \quad (1)$$

$$Y_2 = X'\beta_2 + \varepsilon_2, \quad \varepsilon_2 = \alpha_2 U + V_2. \quad (2)$$

The endogeneity has one source: a latent factor U that enters both equations. V_1 and V_2 are idiosyncratic and independent of each other, of U , and of X . Nothing so far is testable and nothing is identified: γ_1 , α_1 , α_2 and the distribution of U can trade off against one another without changing what is observed.

The restriction that closes the model is what the accompanying papers call *linear scale-consistency*:

$$\alpha_1 = \gamma_1 \alpha_2. \quad (3)$$

Read it as a statement about units. One unit of U shifts Y_2 by α_2 , and that shift carries into Y_1 through γ_1 ; equation (3) says the confounder's *direct* effect on Y_1 is the same size as the effect it transmits through the regressor. It is an algebraic consequence of writing the system linearly in a single factor, not an economic hypothesis — but it is a restriction, and the two-indicator model of Section 2.4 exists precisely to test it rather than maintain it.

Write $\hat{\xi}$ and $\hat{\varepsilon}_2$ for the residuals of Y_1 and Y_2 after regressing each on X , and let

$$m_{jk} = E\left[\hat{\xi}^j \hat{\varepsilon}_2^k\right]$$

denote their moments. Every route below is a function of a handful of these. Two quantities recur:

$$\theta = \alpha_2^2 \text{Var}(U), \quad c = \frac{\theta}{\theta + \text{Var}(V_2)} \in [0, 1],$$

θ being the part of the first-stage residual variance due to the confounder and c its share.

2.2 What holds with no distributional assumption

Let $\tilde{\gamma} = m_{11}/m_{02}$ be the least-squares slope on Y_2 in the long regression. Under (3) alone,

$$\tilde{\gamma} = \gamma_1(1 + c), \tag{4}$$

and since $c \in [0, 1]$ by construction,

$$\gamma_1 \in [\tilde{\gamma}/2, \tilde{\gamma}] \quad \text{for } \gamma_1 > 0, \tag{5}$$

with the endpoints exchanged when $\gamma_1 < 0$. Three things follow, and they are the backbone of the command.

1. **The sign is identified.** Least squares cannot reverse the sign of the coefficient under this model.
2. **The direction of the correction is known.** Any consistent estimator must be *smaller in magnitude* than the least-squares coefficient, and no smaller than half of it.
3. **The interval is exactly the feasible set.** $c < 0$ or $c > 1$ means a negative variance. A candidate outside (5) is therefore not a large estimate but an infeasible one, and the command reports it as such.

No distribution has been assumed, and nothing above can fail unless the model itself is wrong. This is why `freeiv` prints the interval on every call, whatever route is requested.

2.3 Buying a point: the routes of the maintained model

To move from the interval to a point, one further assumption is needed. Table 1 lists what each route asks for. The choice among them is substantive, not statistical, and `freeivmenu` exists to say which assumptions the data can carry before any of them is made.

Table 1: The routes of the maintained model and the single assumption each adds to (3)

method()	order used	what it assumes beyond the model
bounds	2	nothing; reports (5) alone
ols	2	nothing; $\tilde{\gamma}$ is the upper endpoint
qme	3	$E[U^3] \neq 0$: the confounder is skewed
sce	2	$\text{Var}(V_1) = \text{Var}(V_2)$
rre	2	$\text{Var}(V_1) = \gamma_1^2 \text{Var}(V_2)$, the scale-free form of the same idea
hme	3	$E[V_2^3] = 0$: a symmetric first-stage disturbance
qbe	2	the calibrated relation between sub-sample bias and quantile
gmm	2,3,4	$E[U^3] \neq 0$, and that the fourth-order identities hold; over-identified, so testable
pgmm	2,3,4	the same, with the second- and third-order conditions imposed exactly

Third order: the quadratic moment estimator

The default route. Under (3), the third-order moments satisfy a quadratic in γ_1 ,

$$2m_{03}\gamma_1^2 - 3m_{12}\gamma_1 + m_{21} = 0, \quad (6)$$

whose discriminant is

$$D = 9m_{12}^2 - 8m_{03}m_{21} = \gamma_1^2 (2A - B)^2, \quad A = \alpha_2^3 E[U^3], \quad B = E[V_2^3].$$

The only requirement is $E[U^3] \neq 0$. The form of D matters in practice: it vanishes when $B = 2A$, and the estimator's standard error is proportional to $1/\sqrt{D}$, so it explodes as that surface is approached. **freeiv** therefore reports D with its standard error and a z against zero, and when $D < 0$ it says that no real root exists and falls back on the vertex rather than returning a number without comment. Of the two roots, the one inside (5) is retained; the output states how many lie there. See Araar (2026c).

Second order with a variance restriction

Two routes close the system at second order by restricting the idiosyncratic variances. **sce** assumes $\text{Var}(V_1) = \text{Var}(V_2)$; **rre** assumes $\text{Var}(V_1) = \gamma_1^2 \text{Var}(V_2)$. The distinction is not cosmetic. The first is a restriction on the *units* of Y_1 and Y_2 — rescaling the dependent variable breaks it — while the second is scale free and is therefore a restriction on the model. The command reports both $k = \sigma_{V_1}^2 / \sigma_{V_2}^2$ and $k^* = \sigma_{V_1}^2 / (\gamma_1^2 \sigma_{V_2}^2)$, implied by whatever value was retained, and warns when k^* is far from one. Both closed forms, and the cube root of the next subsection, are derived in Araar (2026a).

Symmetry

`hme` identifies γ_1 from a cube root under the assumption that the first-stage disturbance is symmetric, $E[V_2^3] = 0$. It is the cheapest route when it applies and the most dangerous when it does not, because it returns a comfortable-looking number either way. `freeivmenu` reports $B = E[V_2^3]$ directly, which is the statistic that tells the two cases apart.

Quantile bias extrapolation

`qbe` estimates the bias on sub-samples ordered by the first-stage residual and extrapolates to the bias-free point through a relation calibrated on the maintained model. It uses only second-order moments and requires no optimisation, but the calibration was fitted on skewed confounders, and the command flags the cases where the data fall outside that domain. It is not smooth, so it carries no analytic standard error; use the `bootstrap` prefix.

Fourth order: two GMM routes and a specification test

Adding the fourth-order moments gives nine moment conditions $h_1, \dots, h_9$ in eight free parameters $(\gamma_1, \theta, \sigma_{V_1}^2, \sigma_{V_2}^2, A, B, A_4, B_4)$, hence one over-identifying restriction and a Hansen J on one degree of freedom. This is the only route in the package that can test the maintained model from a single endogenous regressor. See Araar (2026b).

`method(gmm)` minimises the criterion *deterministically*, and the reason is worth a paragraph because it is easy to get wrong. Fix γ_1 and θ : the only non-linearity left in h is the product $\theta\sigma_{V_2}^2$, and fixing θ removes it, so all nine residuals become linear in the six remaining parameters. The eight-parameter problem is therefore a two-dimensional surface in (γ_1, θ) with a bounded linear least-squares step at each node, and a surface can be gridded. There is no seed and no starting value: the same data give the same number. Multi-start local optimisation over the eight parameters does not have that property — on most of the datasets of Araar (2026b) it returns a *local* minimum.

Because the criterion is often nearly flat, the route reports more than a point. Profiling J over γ_1 and collecting

$$\mathcal{C} = \{\gamma_1 \in [\tilde{\gamma}/2, \tilde{\gamma}] : J(\gamma_1) - \min J \leq 3.84\} \quad (7)$$

gives an Anderson–Rubin-type region. It inverts a test of the moment conditions at each fixed γ_1 rather than inverting a Wald statistic around the optimum, so it stays valid where a standard error does not: on the boundary of the parameter space, and on a flat criterion. The command prints $\mathcal{C}$ and its share of the identified interval. **That share is the statistic to read first.** When it is well below one, the higher-order moments are doing real work; when it fills the interval, they have added nothing to (5) and the point estimate is the argmin of a flat surface.

`method(pgmm)` imposes $h_1, \dots, h_5$ exactly, which solves $\theta, \sigma_{V_1}^2, \sigma_{V_2}^2, A, B$ in closed

form at each γ_1 and leaves four residuals in three unknowns — again one over-identifying restriction. This is a *different* estimator, not the same one solved differently: it weights the first five conditions infinitely rather than optimally. It is kept for two reasons. It is one-dimensional, hence gridded exhaustively and never troubled by local minima. And substituting A into its fifth residual returns $m_{21} - 3\gamma_1 m_{12} + 2\gamma_1^2 m_{03}$, which is the quadratic (6) itself — the profiled GMM exhibits the third-order route as the part the fourth moments reweight. A wide gap between the two is a symptom of a flat criterion, not an error in either.

What the J tests, and when it can test anything. It tests (3) jointly with the linearity of the confounder’s effect; a rejection does not say which failed. And it cannot always test: writing the unrestricted model with the direct loading free, τ in place of $2\gamma_1$, the Jacobian of the nine moments has determinant

$$-(\gamma_1 - \tau)^5 (B \kappa_U - A \kappa_{V_2}),$$

so the restriction is testable everywhere except where the second factor vanishes — and a normal V_2 sits exactly there, since it makes both B and κ_{V_2} zero. **freeiv** reports that factor with a z , and says explicitly when a non-rejection carries no information because the design sits near that surface.

2.4 Two indicators of one confounder

When a second endogenous variable loads on the same latent factor, the model becomes

$$Y_1 = X'\beta_1 + \gamma_2 Y_2 + \gamma_3 Y_3 + a_1 U + V_1, \quad (8)$$

$$Y_2 = X'\beta_2 + a_2 U + V_2, \quad Y_3 = X'\beta_3 + a_3 U + V_3, \quad (9)$$

and the situation changes qualitatively: a_1 is *free* and identified from second- and third-order cross-moments, with no restriction tying it to γ_2 and γ_3 . Scale consistency is then a hypothesis the data can refute rather than a maintained assumption, and **freeiv** reports the test as $a_1 - (\gamma_2 a_2 + \gamma_3 a_3)$ with a standard error. See Araar (2026d).

Note carefully what Y_3 is and is not. It is *not* an instrument: it enters the outcome equation, and the command prints what it would have returned had it been used as one, precisely so the difference is visible. What it must be is a second observable driven by the same factor.

Three guards protect the closed form, from Proposition 1 of that paper. They fire when the third-order moments imply something no one-factor structure can produce: opposite signs of $E[e_2^2 e_3]$ and $E[e_2 e_3^2]$; an incoherent product and ratio of the loadings; or a negative implied variance. When a guard fires the closed form is not reported at all, and **e(guard)** records which one. A refusal is information about the candidate indicator, not a failure of the command.

2.5 The literature, on the same data

Seven further routes are computed for comparison. They do not share the maintained model. Six of them are reported below a separating line, with a note when they fall outside (5); **lsz** is reported above it, among the routes of the maintained model, because its parameter space contains theirs — what differs is the identifying assumption, not the system.

Table 2: The comparison routes and what each maintains instead

<code>method()</code>	identifying assumption
lewbel12	heteroscedasticity of the first-stage error in the controls, which generates internal instruments
lsz	the same triangular system with the direct loading <i>free</i> ; parameter spaces nest, identifying assumptions do not
copula	a Gaussian copula between the endogenous regressor and the structural error (Park and Gupta 2012)
rank	a control function in the rank of the first-stage residual
rpiv	residual purging; inconsistent in general, reported because it is used
ape	an adaptive rule between least squares and rpiv
oster	proportional selection on observables and unobservables, with δ and $R_{\max}$ set by the user

lewbel12 follows Lewbel (2012) as **ivreg2h** (Baum and Schaffer 2012) implements it, **lsz** follows Lewbel et al. (2024) as **trigmm** does, and **oster** follows Oster (2019) as **psacalc** does. Each was validated against its reference command; the agreement is reproduced by the test files distributed with the package.

3 The command

3.1 Syntax

One endogenous regressor:

```
freeiv depvar [indepvars](endogvar) [if][in][weight][, options]
```

Two endogenous regressors sharing one latent confounder:

```
freeiv depvar [indepvars](endogvar1 endogvar2) [if][in][weight][, level(#)  
noheader]
```

The second model is selected by the syntax alone — two variables inside the parentheses — and needs no method name. **fweights**, **pweights** and **aweight**s are allowed. *indepvars* may contain factor variables; they are expanded with the base level omitted, and the estimate is identical to one obtained from indicators built by hand, since drop-

ping any one level of a full set spans the same column space. The `bootstrap`, `jackknife` and `svy` prefixes work: `e(b)` and `e(V)` are posted whenever an estimate exists.

3.2 Options

`method(name)` selects the route; the default is `method(qbe)`. *name* is one of `bounds`, `ols`, `qme`, `sce`, `rre`, `hme`, `qbe`, `gmm`, `pgmm`, `lsz`, `lewb12`, `copula`, `rank`, `rpiv`, `ape`, `oster`, or `all`.

`quantile(#)` sub-sample quantile used by `method(qbe)`; default `quantile(0.25)`.

`delta(#)` selection parameter of `method(oster)`; default `delta(1)`.

`rmax(#)` R^2 ceiling of `method(oster)`; default `min(1.3R_1, 1)`.

`sign(#)` orientation of `method(lsz)`, 1 or -1 ; default `sign(1)`. It selects which branch the solver follows, as `trigmm`'s own option does.

`level(#)` confidence level; default `level(95)`.

`noheader` suppresses the header.

3.3 Which option acts on which route

An option that is legal but inert is the same trap as one silently absorbed, so it is worth stating plainly: `quantile()` acts on `qbe` alone, `delta()` and `rmax()` on `oster` alone, `sign()` on `lsz` alone. Every other route takes none of them, the default included. Under `method(all)` all four act, each on its own row of the table.

Seven point estimators — `qbe`, `lewb12`, `copula`, `rank`, `rpiv`, `ape`, and `oster` — have no analytic standard error, so no confidence interval is printed for them, and `bounds` returns no point at all. `level()` is not disabled there, because under a `bootstrap`: prefix the prefix owns the coefficient table and uses `level()` itself.

3.4 What the command refuses

`freeiv` declines rather than guesses. It refuses `method(lewb12)` and `method(oster)` when no exogenous control is supplied, since both need one. It refuses an option that steers a route the two-indicator model does not use. It refuses `sign()` outside $\{-1, 1\}$ and `rmax()` outside $(0, 1]$. And, in the two-indicator model, it refuses to report the closed form when a guard of Proposition 1 fires.

3.5 How to read the output

The display has four blocks, and they are meant to be read in order.

The coefficient with its standard error and interval when one is valid. When it is not — a non-smooth route, a minimum on the boundary, a solver that did not converge — the point is printed alone and a note says which of those it is. A suppressed standard error is a statement, not an omission.

Identification the least-squares slope, the interval (5), the skewness of the first-stage residual, the discriminant D with its standard error and a z against zero, the two roots of (6) and how many lie inside the interval.

Nuisances implied by the retained value θ , $\sigma_{V_2}^2$, $\sigma_{V_1}^2$, and the ratios k and k^* , computed *from the coefficient actually retained*. A negative entry here is the loudest signal the command produces: it says the retained value lies outside the region the model allows, so the numbers above it are arithmetic rather than estimates.

Over-identification at order four the joint GMM, its J , the region $\mathcal{C}$ and its share of the interval, then the profiled GMM and its J , then the factor that decides whether that J can test anything at all.

3.6 Stored results

`freeiv` is an `e()`-class command; `ereturn list` shows the full set. The ones most often wanted are `e(gamma)` and `e(se)` for the retained estimate; `e(lo)` and `e(hi)` for the interval; `e(ols)`, `e(qme)`, `e(sce)`, `e(rre)`, `e(hme)` for the closed forms; `e(g_jgmm)`, `e(J9)`, `e(p_J9)` for the joint GMM, with `e(ar_lo)`, `e(ar_hi)` and `e(ar_frac)` for the region (7) and `e(j_bound)`, `e(j_weak)` for its flags; `e(g_gmm)`, `e(J)`, `e(p_J)` for the profiled GMM; `e(disc)` and `e(disc.z)` for the discriminant; and `e(theta)`, `e(sV1)`, `e(sV2)` for the implied nuisances. In the two-indicator model, `e(g2)`, `e(g3)` and `e(a1)` are the structural coefficients and the free loading, `e(sc_d)` and `e(z_sc)` the scale-consistency test, and `e(guard)` records which guard fired, or zero.

3.7 Companion commands

`freeivmenu` reports, *before* any estimation, which assumptions the data can carry: the width of the interval, the skewness of the first-stage residual with a z , the discriminant and the standard error it implies for the quadratic route, $B = E[V_2^3]$ for the symmetry route, k and k^* for the variance restrictions, and a heteroscedasticity test for `lewbel12`. Each line carries a verdict in words.

`freeivdiag` takes any value of γ — from `freeiv`, from two-stage least squares, from another paper — and reports what it implies about the unobservables under the maintained model: the implied variances, the confounding share, and whether the value is feasible at all.

`freeivtest` reports an endogeneity test, the agreement between routes, and an outside estimate against the identified set.

`freeivreport` assembles all of it into one table, optionally written to a CSV.

4 Examples

Four datasets are distributed with the package and retrieved with `net get freeiv`. Three are simulated, and their generating equations are printed in the help file, so none is a black box.

4.1 Simulated data where the model holds

`freeiv.sim1.dta` is generated from

$$\begin{aligned} u &= (\chi_3^2 - 3)/\sqrt{6}, & v_1, v_2, x &\sim N(0, 1), \\ y_2 &= 0.30x + 0.80u + v_2, \\ y_1 &= 0.60x + 0.40y_2 + 0.32u + v_1, \end{aligned}$$

so the truth is $\gamma_1 = 0.40$ and scale consistency holds by construction, $0.40 \times 0.80 = 0.32$. The confounder is skewed, so the third-order routes are available.

```
. use freeiv_sim1, clear
. freeivmenu y1 x (y2)
. freeiv y1 x (y2), method(all)
```

Read the menu first, then every route against the interval and against 0.40. On these data the region (7) covers about 64% of the identified interval: identification is strong, and the moments of orders three and four genuinely narrow what the second-order algebra gives on its own. Keep that figure in mind for Section 4.3.

4.2 The same data at the edge of identification

`freeiv.sim2.dta` differs in one respect: v_2 is drawn from the same skewed law as u instead of a normal. Scale consistency still holds and the truth is still 0.40 — yet the quadratic route returns nothing.

The reason is exact rather than accidental. A standardised χ_3^2 has third moment $24/6^{3/2} = 4/\sqrt{6} = 1.6330$, so with $\alpha_2 = 0.80$ and U and V_2 drawn from that same law, $A = 0.8^3 \times 1.6330 = 0.8361$ and $B = 1.6330$. Then $2A - B = 0.0392$ rather than zero, and the population discriminant $\gamma_1^2(2A - B)^2$ is 0.00025, against 0.447 in `freeiv.sim1`, where V_2 is normal and B is zero. The design sits almost exactly on the surface where the two roots merge, and in any one sample D lands on either side of zero.

What the output shows is worth more than an estimate. The quadratic route reports no real root and falls back on the vertex, saying so, and its standard error is enormous. `method(hme)` meanwhile returns a comfortable-looking number near 0.34 — while the

very assumption it rests on, $E[V_2^3] = 0$, is violated by construction. A route that fails loudly is safer than one that fails quietly, and the menu's B is what tells them apart.

4.3 Card: one endogenous regressor

`freeiv_card.dta` is the Card (1995) extract used in the empirical section of Araar (2026d): log wage on schooling, with experience and indicators for race, region and metropolitan status. The menu comes first.

```
. use freeiv_card, clear

. freeivmenu lwage exper black south smsa (educ)

Identification menu: what these data can carry
lwage on educ, controls  exper black south smsa      n =      3010
-----
```

route	signal	value	verdict
bounds	width of the interval [gamma-tilde/2, gamma-tilde]	0.0369 [0.0369, 0.0738]	always
third order: qme,	skewness of eps2	0.1666	
hme, lsz	z of m03	2.38	weak
	z of the discriminant D	0.27	absent
	implied s.e. of the QME	0.0469	wide
	A = alpha2^3 E[U^3]	1.3621	
	mu = A/(A+B)	1.1097	
equal variances: rre	k* = sV1/(g^2 sV2), sets 1	10.4121	doubtful
	rre = m20/(2 m11)	0.2910	outside
and its sce variant	k = sV1/sV2, sce sets 1	0.0433	doubtful
symmetry of V2: hme	B = E[V2^3], hme sets 0	-0.1346	plausible
heteroskedasticity:	F of eps2^2 on X	8.06	present
lewbel12	p	0.0000	
two indicators	freeiv y1 x (y2 y3)	.	by syntax

```
-----
```

The verdicts are already discouraging for a point estimate: the third-order signal is weak, and the discriminant is indistinguishable from zero, which is where the quadratic route's standard error explodes. Running every route confirms it and adds two diagnostics the menu cannot give.

```
. freeiv lwage exper black south smsa (educ), method(all)

Estimates of gamma      estimate      std. err.
ols                     0.073807      0.003646
qme                     0.064462      0.046876
sce                     0.037608      0.001857
rre                     0.291008      0.014791
hme                     .
qbe                     0.079241      .
gmm                     0.073806      .      J  0.841 p 0.359
pgmm                    0.073707      0.029984      J  0.844 p 0.358
lsz                     0.073284      .      did not converge
bounds [0.036904, 0.073807]
the literature, on the same data
lewbel12                0.084535      outside the bounds
copula                  0.018892      outside the bounds
```

```

rank          0.110185      outside the bounds
rpiv          0.039402
ape           0.039402
oster        0.083893      outside the bounds
an estimate outside the bounds implies a negative variance
under scale consistency; -freeivdiag, gamma(#)- says which

```

Four independent diagnostics say the same thing. The discriminant carries almost no signal. The region (7) covers 91% of the identified interval — all but a sliver of what the assumption-free bounds admit is a value the nine moment conditions cannot reject — against 64% on the simulated data, so the fourth-order moments add almost nothing here; add `expersq` to the controls and the share is 100%. The joint GMM's minimum sits on the boundary with $\hat{\theta}$ at its floor, and no standard error is printed because none is valid there. And the LSZ solver does not converge, which is reported rather than hidden behind what would otherwise be the tightest interval in the table.

```

Over-identification at order four
joint nine-moment GMM, method(gmm)
  gamma          0.073806
  J, 1 df        0.8407   P>chi2 0.3592
  J within 3.84 of its minimum on [0.040287, 0.073806]
  that is 90.8% of [gamma-tilde/2, gamma-tilde]
note: the minimum is ON THE BOUNDARY -- gamma at an end of
      the interval, or theta at its floor. Neither the
      standard error nor the chi2(1) law for J is valid
      there, so no standard error is printed.
note: theta < 0.05 at the minimum: the implied confounder
      has almost no variance, so kurt_U is not defined and
      there is nothing for the higher moments to bind on.
profiled GMM, method(pgmm)
  gamma          0.073707
  J, 1 df        0.8443   P>chi2 0.3582

```

Four of the six comparison routes fall outside the interval. That is not a verdict on those estimators: it says that their maintained models and this one disagree about these data, and under *this* model those values imply a negative variance. `freeivdiag`, `gamma(0.084535)` says which one.

The honest reading of Card with a single endogenous regressor is an interval, 0.037 to 0.074, and not a point. This is the ordinary case, and it is the reason `freeivmenu` exists.

4.4 Card with a second indicator of the confounder

Mother's education loads on the same family background that confounds schooling, and it is itself endogenous in the wage equation. It is not an instrument — and using it as one is precisely the comparison the command prints.

```
. freeiv lwage exper expersq black south smsa (educ motheduc)
```

Now the coefficient is identified as a point, with a standard error, and the confounder's own loading a_1 is free and estimated rather than restricted. These lines re-

produce Table 6b of Araar (2026d) on that paper’s own data: $\gamma_2 = 0.066$, $\gamma_3 = -0.009$, $a_1 = 0.067$, an instrument gap of 0.102 with `motheduc`, and $\gamma_2 = 0.035$ with `fatheduc`, the departure from scale consistency being 0.032 and 0.563 respectively. The agreement is checked rather than asserted: a test file distributed with the package runs that paper’s own script and compares value by value.

Two candidate indicators from the same file are *refused* by the guards, which is the intended behaviour and worth seeing.

```
. freeiv lwage exper expersq black south smsa (educ IQ)
. freeiv lwage exper expersq black south smsa (educ KWW)
```

KWW is refused because $E[e_2^2 e_3]$ and $E[e_2 e_3^2]$ have opposite signs, IQ because an implied variance is negative — reason by reason, Table 6a of that paper.

5 Discussion

Three habits are worth stating, because they are what the command is designed to support.

Read the interval before the point. It costs no distributional assumption and cannot fail unless the model itself is wrong. Everything else in the output is an attempt to narrow it, and the attempt can fail.

Read the diagnostic before the estimate. `freeivmenu` says which assumptions the data carry; the discriminant says whether the quadratic route has anything to work with; the share of the interval covered by (7) says whether the fourth-order moments added anything. When those say no, the point estimate is an artefact of a flat criterion, and the command declines to dress it in a standard error.

Read the routes together. Agreement between estimators with different maintained models is evidence; disagreement locates the assumption that is doing the work. This is why `method(all)` exists and why the comparison routes are computed on the same data rather than cited.

The limitations are equally plain. The model has one latent confounder and is linear in it; with two factors the identification arguments do not hold, and a second factor that is Gaussian is undetectable at any order of moments. The J tests scale consistency and linearity jointly and cannot separate them. And an unmodelled interaction between the regressor and the confounder biases the estimator in a window where the J remains silent. The two-indicator model of Section 2.4 addresses the first of these by freeing the direct loading, which is why a second observable on the same factor is worth more than an extra control.

What the third order cannot tell apart. One caveat deserves its own paragraph, because it concerns every instrument-free route and not only these. In the single-indicator model the three third-order moments serve three unknowns, (γ_1, A, B) : the system is just-identified, so any triple (m_{03}, m_{12}, m_{21}) can be rationalised by a con-

founder, and a mechanism with no confounder at all can produce the same triple. Take $Y_1 = aY_2 + bY_2^2 + e$ with Y_2 standard normal, e independent, and no latent factor. Then $m_{03} = 0$, $m_{12} = 2b$, $m_{21} = 4ab$, and (6) reduces to $-6b\gamma_1 + 4ab = 0$: the route returns $\gamma_1 = 2a/3$ for every $b \neq 0$ — a correction of one third of the least-squares slope that does not depend on the size of the quadratic term, lies inside (5), carries a positive discriminant, and implies a confounder share $c = 1/2$ with positive variances. Nothing at order three can object, because nothing at order three is over-identified. The fourth order can: on this design with $b = 0.1$ and $n = 20,000$ the J of `method(pgmm)` is 14 and that of `method(gmm)` is 5.4, both rejecting. But the power of that test falls with b^2 and with n , while the bias does not: at $b = 0.02$ the J is 0.2 and the estimate is still 0.70. Three things follow. The z of the discriminant in `freeivmenu` measures the strength of a third-order signal, not its source — a curvature in $E[\hat{\xi} \mid \hat{\varepsilon}_2]$ is what a skewed confounder and a non-linear outcome equation both produce, and a RESET-type check on $\hat{\varepsilon}_2^2$ rejects under either. The two-indicator model is the least exposed route in the family: its loading layer never reads Y_1 , so a non-linearity of the outcome equation can reach only the causal layer, where γ_2 is obtained by two routes whose disagreement detects it. And a valid external instrument, where one exists, is the only thing that identifies γ_1 with no assumption on the form of the outcome equation at all — which is why an instrument-free route is a complement to one, never a substitute.

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The reference implementations `ivreg2h`, `trigmm`, and `psacalc` were used to validate the comparison routes.

7 References

- Araar, A. 2026a. Correcting endogeneity without external instruments: Four closed-form estimators for the linear structural model. Working paper, Zenodo. <https://doi.org/10.5281/zenodo.22067980>.
- . 2026b. Instrument-free estimation under linear scale-consistency: Closed-form identification, partial bounds, and higher-moment GMM. Working paper, Zenodo. <https://doi.org/10.5281/zenodo.22753299>.
- . 2026c. The quadratic moment estimator for instrument-free endogeneity correction. Working paper, Zenodo. <https://doi.org/10.5281/zenodo.22068143>.
- . 2026d. Two indicators of one latent confounder: Closed-form identification of the triangular model with a free proxy effect. Working paper, Zenodo. <https://doi.org/10.5281/zenodo.22764001>.
- Baum, C. F., and M. E. Schaffer. 2012. `ivreg2h`: Stata module to perform instrumental variables estimation using heteroskedasticity-based instruments. Statistical Software Components S457555, Boston College Department of Economics.

- Card, D. 1995. Using geographic variation in college proximity to estimate the return to schooling. In *Aspects of Labour Market Behaviour: Essays in Honour of John Vanderkamp*, ed. L. N. Christofides, E. K. Grant, and R. Swidinsky, 201–222. Toronto: University of Toronto Press.
- Lewbel, A. 2012. Using heteroscedasticity to identify and estimate mismeasured and endogenous regressor models. *Journal of Business and Economic Statistics* 30: 67–80.
- Lewbel, A., S. M. Schennach, and L. Zhang. 2024. Identification of a triangular two equation system without instruments. *Journal of Business and Economic Statistics* 42: 14–25.
- Oster, E. 2019. Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business and Economic Statistics* 37: 187–204.
- Park, S., and S. Gupta. 2012. Handling endogenous regressors by joint estimation using copulas. *Marketing Science* 31: 567–586.

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